# The Price of Position: Does Listing Order Shape the Diffusion of AI Research?

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**Abstract**

Every day at 2:00 PM Eastern, arXiv draws a line: papers submitted before appear at the bottom of today's announcement; those after appear at the top of tomorrow's. I exploit this cutoff to estimate the causal effect of listing position on AI research diffusion. Using a regression discontinuity design on over 8,000 machine learning papers (2012–2020), I find that crossing the cutoff shifts position from the 80th to the 10th percentile. Despite this massive visibility shock, I find no evidence of large citation effects at any horizon — the design rules out effects exceeding 1 log point (170%) but cannot detect moderate effects (20–30%) found elsewhere. Point estimates are negative but insignificant. The design passes all validity tests. These results suggest that descriptive position-citation correlations on arXiv reflect author selection rather than causal visibility effects.

**JEL Codes:** O31, O33, I23, D83

**Keywords:** science of science, attention, arXiv, citation premium, regression discontinuity, AI research, platform design

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## 1. Introduction

On June 12, 2017, a team of eight Google researchers submitted "Attention Is All You Need" to arXiv. The paper introduced the Transformer architecture, which would become the foundation of GPT, BERT, and every large language model that followed. As of 2025, it has over 130,000 citations and underpins products used by billions of people. But had Vaswani and colleagues submitted their paper twelve minutes later — after arXiv's daily cutoff — it would have appeared first in the next day's announcement list instead of buried among that day's 50 other machine learning uploads. Would that have mattered?

This paper asks whether the arbitrary visibility that arXiv's platform design confers on some papers over others has lasting consequences for the diffusion of scientific ideas. The question matters because arXiv has become the central marketplace for AI research: over 200,000 machine learning papers were posted between 2012 and 2020, and the site's daily announcement email is the primary channel through which researchers, journalists, and industry labs discover new work. If listing position causally affects which papers get read, cited, and adopted, then a website's default sort order is shaping the trajectory of a $100 billion industry.

I exploit a sharp institutional rule that generates quasi-random variation in visibility. Every weekday at exactly 2:00 PM Eastern Time, arXiv closes its submission window for the current day's announcement.[1] Papers submitted before the cutoff are included in today's listing, ordered chronologically from earliest to latest submission. Papers submitted after the cutoff wait for tomorrow's listing, where they appear first — at the top of the list. This creates a discontinuity: a paper submitted at 1:55 PM is listed near the *bottom* of today's batch (position 50 out of 60), while a paper submitted at 2:05 PM is listed near the *top* of tomorrow's batch (position 3 out of 65). Within a narrow window around the cutoff, which side a paper lands on is determined by idiosyncratic factors — network latency, file preparation time, the length of a coffee break — that are plausibly unrelated to paper quality.

I implement a regression discontinuity design (RDD) using this cutoff. The running variable is the number of minutes between submission time and the 14:00 ET deadline. The treatment is the bundle of visibility changes that come from crossing the cutoff: a dramatic improvement in listing position, offset by a 24-hour delay in announcement. The net effect captures the market's revealed preference for position versus timeliness — a fundamental parameter for platform design.

The data combine two sources. First, I collect arXiv metadata with exact submission

[1] The cutoff has been at 14:00 ET throughout the sample period. See https://info.arxiv.org/help/availability.html.

timestamps (hour, minute, second) for all papers in core AI/ML categories (cs.AI, cs.CL, cs.LG, stat.ML, cs.CV) submitted between 2012 and 2020. Second, I match these papers to OpenAlex (Priem et al., 2022), which provides citation counts by year, author institutional affiliations, and publication venue information. This matching allows me to trace each paper's diffusion trajectory from initial posting through long-run citation accumulation and adoption by industry research laboratories.

Crossing the cutoff causes a massive improvement in listing position: the average paper moves from approximately the 80th percentile to the 10th percentile of within-batch ordering, with a robust RDD estimate of $-0.70$ on the position percentile ($z = -17.1$, $p < 0.001$). This first stage is sharp and unambiguous. Second — and this is the paper's central surprise — this large visibility shock does *not* translate into detectably more citations. Point estimates for log citations at all horizons (1-year, 3-year, 5-year, total) are negative and statistically insignificant: the 3-year estimate is $-1.086$ log points ($p = 0.14$). The design rules out large positive effects (greater than approximately 170% of baseline citations) but cannot detect moderate effects of 20–30% as found by Feenberg et al. (2017). Third, industry adoption shows a similarly null point estimate; top-venue publication could not be estimated due to data missingness. Fourth, the design passes all standard validity tests: the McCrary density test shows no significant bunching ($p = 0.25$), pre-determined covariates are smooth through the cutoff, and placebo cutoffs produce null results.

These results are robust to extensive sensitivity analysis. Varying bandwidths from 50 to 200 percent of the MSE-optimal choice produces estimates that range from moderately positive to moderately negative, with none achieving statistical significance at the 5 percent level. Donut RDD specifications that exclude papers within $\pm 5$ minutes of the cutoff produce similar estimates. Alternative polynomial orders (quadratic, cubic), kernel functions (Epanechnikov, uniform), and sample restrictions (excluding conference deadline months) all confirm the absence of detectable effects.

This paper contributes to several literatures. Most directly, it advances the *science of science* (Fortunato et al., 2018; Sinatra et al., 2016; Wang et al., 2013) by providing causal evidence — in this case, a well-identified null on large effects — on the relationship between platform visibility and knowledge diffusion. The closest antecedent is Feenberg et al. (2017), who exploit the random ordering of NBER's weekly digest to show that first-listed working papers receive 30 percent more downloads and citations. My design differs in three ways: arXiv has *not* randomized its listing order (making the positional effect actively exploitable), I focus on the explosive CS/AI literature rather than economics, and the treatment bundles position improvement with a 24-hour delay — creating a clean test of whether position *alone* drives the correlations documented descriptively. I also build on work documenting the

"first-listed advantage" on arXiv in physics (Dietrich, 2008; Haque and Ginsparg, 2009, 2010), providing the first formal causal design and the first analysis outside physics. The absence of large effects suggests that the descriptive position-citation correlation in prior work reflects selection (strategic authors are better) rather than a large causal visibility effect, though moderate effects consistent with Feenberg et al. (2017) cannot be ruled out.

Second, the paper speaks to the economics of attention and information overload (Jones, 2009; Bloom et al., 2020). The surprising finding is that timeliness — being included in today's announcement rather than tomorrow's — appears to matter at least as much as position within a given announcement. This is inconsistent with simple models where preferential attachment (Barabási and Albert, 1999; Merton, 1968) is the dominant force, and suggests that the scientific marketplace operates more efficiently than pessimists fear: good ideas find their audience regardless of where they appear in a list. This connects to the broader literature on how institutional design shapes the direction of innovation (Li, 2017; Hill and Stein, 2021; Azoulay et al., 2014).

Third, these findings have direct implications for *platform design in science*. ArXiv serves over 2 million papers and 300,000 daily users. The absence of large detectable effects from a 70 percentage point position shock suggests that randomizing announcement order — as NBER did following Feenberg et al. (2017) — may be less consequential than previously believed, at least in the AI/ML literature where alternative discovery channels abound. More broadly, the results caution against extrapolating from descriptive position-outcome correlations to causal effects: what appears to be platform influence may be driven entirely by selection.

Section 2 describes arXiv's institutional setting. Section 3 presents a conceptual framework. Section 4 describes the data, Section 5 the empirical strategy, and Section 6 the results. Section 7 discusses mechanisms and external validity. Section 8 concludes.

## 2. Institutional Background

### 2.1 arXiv and the Distribution of Scientific Knowledge

ArXiv is the world's largest open-access preprint repository. Founded in 1991 by Paul Ginsparg at Los Alamos National Laboratory (Ginsparg, 2011), it now hosts over 2.4 million papers across physics, mathematics, computer science, quantitative biology, and related fields. In machine learning and artificial intelligence — the categories studied in this paper — arXiv has become the de facto publication venue: virtually every significant AI paper is posted to arXiv, often months before formal journal or conference publication.

The platform's importance for AI research cannot be overstated. The Transformer paper (Vaswani et al., 2017), BERT (Devlin et al., 2019), GPT-3 (Brown et al., 2020), ResNet (He

et al., 2016), and Generative Adversarial Networks (Goodfellow et al., 2014) were all first disseminated through arXiv. Industry research labs at Google, Meta, OpenAI, and Microsoft treat arXiv postings as the primary signal of research output. Hiring committees, grant reviewers, and journalists use arXiv to identify emerging researchers and breakthrough results. In this ecosystem, when and where a paper appears in the daily listing is not a trivial detail — it determines who sees it first.

### 2.2 The Daily Submission Cycle

ArXiv operates on a strict daily batch cycle, illustrated in Figure 1. The key institutional feature is a **daily submission cutoff at 14:00 Eastern Time** (2:00 PM ET), active Monday through Friday.

**arXiv Daily Cycle (All Times Eastern)**

*Monday–Thursday:*

- 14:00 ET: Submission window closes
- 14:00–14:00 (next day): Papers accumulate for next batch
- 20:00 ET: Announcement published on website and email

*Thursday–Friday:*

- Thursday 14:00: Window closes
- Sunday 20:00: Announcement published (stays on front page 3 days)

*Friday–Monday:*

- Friday 14:00: Window closes
- Monday 20:00: Announcement published

**Figure 1:** The arXiv Daily Submission and Announcement Cycle

Papers submitted *before* the 14:00 cutoff on a given day are included in that day's announcement, published at 20:00 ET. Papers submitted *after* the cutoff are held for the next business day's announcement. Within each announcement batch, papers are ordered strictly by submission time: the earliest submission appears first, and the latest submission appears last.

This creates a sharp discontinuity. A paper submitted at 13:55 ET is among the *last* in today's batch — having been preceded by every paper submitted since the previous cutoff. A paper submitted at 14:05 ET is among the *first* in tomorrow's batch, preceded only by papers submitted in the minutes after the cutoff closed. The jump in listing position is enormous: from approximately the 80th–90th percentile of the batch to the 1st–10th percentile.

### 2.3 Why Listing Position Matters

Three features of arXiv's design amplify the importance of listing position:

**Email announcements.** ArXiv sends daily email digests to subscribers, listing new papers in each category in submission order. Subscribers in cs.AI or cs.LG receive lists of 50–100 new papers per day. Behavioral research on email engagement shows that items near the top of lists receive disproportionate attention: click-through rates decline monotonically with position (Feenberg et al., 2017).

**Website "new submissions" page.** ArXiv's website displays new submissions in chronological order. The first five papers are visible without scrolling on most screen sizes. Papers below the fold receive substantially less attention, analogous to Google search results where the first page captures over 90 percent of clicks.

**Aggregator and blog coverage.** Third-party services (Papers With Code, Hacker News, Twitter/X bots, Google Scholar alerts) often highlight the first few papers in each day's listing. Being first-listed increases the probability of secondary dissemination through these channels.

### 2.4 The Cutoff as a Natural Experiment

The 14:00 ET cutoff creates variation that is useful for causal inference because, within a narrow window around the deadline, which side a paper lands on is plausibly as-good-as-random. Several factors contribute to this local randomization:

*Network and upload variability.* LaTeX compilation, file conversion, and upload times vary by seconds to minutes depending on server load, file size, and network conditions. A paper whose author clicked "submit" at 13:58 might not register until 14:01 due to processing lag.

*Timezone confusion.* ArXiv's cutoff is stated in Eastern Time, but researchers work in every timezone. A researcher in California (Pacific Time) might believe they are submitting at "11:00 AM" with ample margin, only to be caught by a DST shift or time zone miscalculation.

*Idiosyncratic scheduling.* Whether a researcher finishes writing at 1:50 PM or 2:10 PM depends on meeting schedules, lunch breaks, and other factors orthogonal to paper quality.

At the same time, the cutoff is well-known, and some researchers strategically time their submissions. The most common strategy is to submit immediately after the cutoff to secure the first position in the next day's listing — a phenomenon documented descriptively by Haque and Ginsparg (2009). This strategic behavior creates bunching just after the cutoff, which I address through donut RDD specifications that exclude the immediate vicinity of the deadline.

## 3. Conceptual Framework

### 3.1 Position, Attention, and Cumulative Advantage

I model the relationship between listing position and long-run citations using a simple attention-based framework. Let paper $i$ be posted on day $t$ in position $p_{it}$ within the daily listing. The paper's initial visibility is a decreasing function of position:

$$V_{it} = g(p_{it}) + \eta_{it}, \quad g'(p) < 0 \tag{1}$$

where $g(\cdot)$ captures the monotonically declining attention profile (top-of-list papers receive more clicks) and $\eta_{it}$ represents random variation in readership.

Initial visibility determines early engagement (reads, bookmarks, shares), which triggers a cumulative advantage process (Merton, 1968; Barabási and Albert, 1999). Let $C_{i,t+s}$ denote citations accumulated by $s$ periods after posting. Citations evolve according to:

$$C_{i,t+s} = C_{i,t+s-1} + h(C_{i,t+s-1}, q_i) + \varepsilon_{i,t+s} \tag{2}$$

where $q_i$ is the paper's intrinsic quality and $h(\cdot)$ is increasing in both arguments. The key feature is that $\partial h/\partial C > 0$: papers with more existing citations attract additional citations (preferential attachment), creating a feedback loop between early attention and long-run impact.

### 3.2 Testable Predictions

This framework generates several predictions:

*Prediction 1: Papers with better listing positions receive more citations.* The direct effect of position on visibility ($g'(p) < 0$) translates into higher initial engagement, which compounds through preferential attachment.

*Prediction 2: The effect is larger for papers where the daily listing is the primary discovery channel.* If researchers in large, fast-moving fields (cs.LG, cs.CL) rely more heavily on daily listings, the position effect should be stronger in these categories than in smaller, more specialized ones.

*Prediction 3: The effect is larger for junior authors.* Established researchers have alternative discovery channels (personal networks, Twitter followings, lab websites). Junior researchers without these channels benefit more from listing-position visibility.

*Prediction 4: The effect compounds over time.* Due to preferential attachment, the citation gap between well-positioned and poorly-positioned papers should widen over the first few

years and then stabilize as the attention decay of the initial shock sets in.

### 3.3 Position vs. Timeliness Tradeoff

The arXiv cutoff creates a distinctive two-dimensional treatment. Papers submitted just after the cutoff receive:

1. A *better* listing position (from ∼80th to ∼5th percentile) — increasing visibility
2. A *delayed* announcement (pushed to the next day) — decreasing timeliness

The RDD estimates the *net* effect of this bundle. A positive estimate implies that position dominates timeliness: the value of being seen first exceeds the cost of being seen 24 hours later. This itself is an informative result about the relative importance of attention versus priority in the marketplace for ideas (Strevens, 2003).

## 4. Data

### 4.1 arXiv Metadata

I obtain metadata for papers submitted to arXiv in the categories cs.AI (Artificial Intelligence), cs.CL (Computation and Language), cs.LG (Machine Learning), stat.ML (Statistics: Machine Learning), and cs.CV (Computer Vision) between January 2012 and December 2020. The metadata is accessed through the arXiv API, which provides exact submission timestamps with second-level precision in the form of ISO 8601 dates (e.g., `2018-04-02T17:58:08Z`). I retrieve a systematic sample of papers across the full time period, queried by category and date range.

For each paper, I record: the arXiv ID, submission timestamp (UTC), title, number of authors, abstract length, listed categories, and number of versions posted. I convert all timestamps from UTC to Eastern Time using the `lubridate` package in R, accounting for daylight saving time transitions.

The raw sample contains approximately 8,440 papers with valid timestamps. Listing position within each announcement batch is computed from all papers observed in that batch; the RDD identification requires only that position is computed consistently for papers near the cutoff on the same day, not that every paper on arXiv is observed. I restrict attention to weekday submissions (Monday–Friday), as weekend submissions follow different batching rules and represent a distinct population of authors. This yields a working sample of 7,412 papers. Of these, 1,820 fall within the ±120-minute RDD window, and 289 of the RDD-window papers match to OpenAlex citation records.

### 4.2 OpenAlex Citation Data

I match arXiv papers to the OpenAlex database (Priem et al., 2022), a comprehensive open catalog of over 250 million scholarly works, using arXiv DOIs (format: `10.48550/arXiv.{id}`). For each matched paper, I obtain:

**Citation counts by year.** OpenAlex provides annual citation counts, allowing me to construct citation trajectories at 1, 3, and 5 years post-submission. This is critical for testing whether the visibility effect compounds over time (Prediction 4). Citation counts were retrieved from OpenAlex in early 2025. The 5-year citation window is fully observed for papers submitted through 2019; for 2020 submissions, the 5-year window is partially incomplete, which may attenuate the 5-year estimates.

**Author affiliations.** For each citing paper, OpenAlex records the institutional affiliation of each author. I use this to identify citations from industry research labs, coding a paper as having "industry adoption" if any citing author is affiliated with one of 20 major technology companies (Google/DeepMind, Meta/FAIR, OpenAlex, Microsoft Research, Amazon, Apple, NVIDIA, Anthropic, and others).

**Publication venue.** OpenAlex records the primary publication venue, allowing me to identify papers that were subsequently published at top machine learning conferences (NeurIPS, ICML, ICLR, AAAI, ACL, CVPR).

The overall match rate between arXiv and OpenAlex is approximately 25 percent of the weekday sample, reflecting incomplete DOI coverage in OpenAlex for older and less-cited papers. Crucially, the match rate does not exhibit a discontinuity at the cutoff (RDD estimate: 0.008, $p = 0.72$), confirming that citation-sample selection is not correlated with treatment status.

### 4.3 Variable Construction

**Running variable.** I compute the running variable as the number of minutes between the paper's submission time (in Eastern Time) and the 14:00 ET cutoff. Negative values indicate submission before the cutoff (paper included in today's batch); positive values indicate submission after the cutoff (paper included in tomorrow's batch).

**Listing position.** Within each announcement batch, I rank papers by submission time. The paper submitted earliest receives position 1; the latest receives the batch's total count. I compute the position percentile (position divided by batch size), which ranges from near zero (first listed) to 1 (last listed), to normalize across batches of different sizes.

**Treatment indicator.** $D_i = \mathbb{I}[\text{run_var}_i > 0]$, indicating submission after the cutoff.

**Citation outcomes.** I construct log-transformed citation counts: $\ln(\text{citations}_s + 1)$ for

$s \in \{1, 3, 5\}$ years and total citations. The log transformation accommodates the heavily right-skewed citation distribution. I also construct binary indicators for being in the top 10 percent of citations within each submission year ("highly cited").

**Industry adoption.** A binary indicator equal to one if the paper has been cited by at least one paper with an author affiliated with a major technology company.

**Top venue publication (exploratory).** A binary indicator equal to one if the paper's primary publication venue (as recorded in OpenAlex) is one of the top-6 ML/AI conferences. Due to substantial missingness in the OpenAlex venue field, this variable could not be used as an RDD outcome and is retained here for completeness only.

### 4.4 Sample Construction and Filtering

The sample construction proceeds through several filters. Starting from all papers in the five target categories submitted between January 2012 and December 2020, I first restrict to weekday submissions (Monday through Friday). Weekend submissions follow distinct batching rules — papers submitted between Friday 14:00 ET and Monday 14:00 ET are all grouped into Monday's announcement — making the running variable less well-defined. This restriction removes approximately 14 percent of submissions.

Next, I exclude papers with missing or malformed timestamps. The arXiv API occasionally returns entries with `<published>` dates that do not include time-of-day information, typically for papers migrated from older systems. These represent less than 1 percent of the sample. I also exclude cross-listings where the paper's primary category falls outside the five target categories but the paper appears as a secondary listing in one of them. This ensures that the sample reflects the population of researchers who actively chose to submit to an AI/ML category.

For the RDD analysis sample, I further restrict to papers submitted within $\pm 120$ minutes of the 14:00 ET cutoff (i.e., between 12:00 and 16:00 ET). This window is wide enough to include substantial variation on both sides of the cutoff while excluding papers submitted at unusual hours (late night, early morning) that may represent a different population. The bandwidth-selection procedure in `rdrobust` further narrows this window to the MSE-optimal bandwidth for each specification.

Finally, I merge the arXiv metadata with OpenAlex citation data. The overall match rate is approximately 25 percent (1,845 of 7,412 weekday submissions), reflecting incomplete DOI coverage in OpenAlex for older and less-cited papers. Unmatched papers are included in all analyses where citations are not the outcome (density tests, position discontinuity, covariate balance) but excluded from citation regressions. Reassuringly, the match rate does not exhibit a discontinuity at the cutoff (RDD estimate: 0.008, $p = 0.72$), confirming that

missingness is not correlated with treatment status.

### 4.5 Summary Statistics

Table 1 presents summary statistics for the OpenAlex-matched sample (1,845 of 7,412 weekday papers with citation data) and the RDD analysis subsample (289 matched papers submitted within ±120 minutes of the cutoff). The median paper in the sample has 5 total citations, the mean has 28, reflecting the extreme right skew typical of citation distributions. Papers have an average of 3.2 authors and are cross-listed in 1.8 arXiv categories. Announcement batches are category-specific: within each category, the median daily batch contains approximately 10 papers, though the mean is higher (approximately 31) due to right-skewed batch sizes in larger categories and later years.

The citation distribution is heavily right-skewed, with a small number of "superstar" papers (above 500 citations) pulling the mean well above the median. This motivates the log transformation used throughout the analysis. It also means that the RDD estimates, which capture local average treatment effects at the cutoff, should be interpreted as effects on the typical paper rather than the exceptional one.

The RDD sample is broadly comparable to the matched sample on observable characteristics, with slightly higher average citations (consistent with the fact that papers submitted during business hours may come from more active researchers) and marginally larger batch sizes (reflecting weekday submission patterns). The key identifying assumption requires smoothness only *at the cutoff*, not comparability between the RDD sample and the broader population.

## 5. Empirical Strategy

### 5.1 Regression Discontinuity Design

I estimate the causal effect of crossing the submission cutoff using a sharp regression discontinuity design (Lee and Lemieux, 2010; Cattaneo et al., 2020b). The estimating equation is:

$$Y_i = \alpha + \tau \cdot \mathbb{I}[T_i > 0] + f(T_i) + \varepsilon_i \tag{3}$$

where $Y_i$ is the outcome (log citations, industry adoption), $T_i$ is the running variable (minutes from cutoff, centered at zero), $\mathbb{I}[T_i > 0]$ is the treatment indicator, and $f(\cdot)$ is a local polynomial fitted separately on each side of the cutoff. The parameter of interest, $\tau$, captures the discontinuous jump in the outcome at the cutoff.

**Table 1:** Summary Statistics

| | Matched Sample | | | RDD Sample (±120 min) | | |
|---|---|---|---|---|---|---|
| Variable | Mean | SD | Median | Mean | SD | Median |
| Total citations | 38.89 | 380.14 | 5.00 | 25.10 | 68.04 | 6.00 |
| 1-year citations | 4.27 | 35.51 | 0.00 | 3.77 | 16.94 | 1.00 |
| 3-year citations | 9.89 | 121.07 | 1.00 | 9.76 | 47.71 | 2.00 |
| 5-year citations | 15.05 | 226.07 | 1.00 | 13.10 | 51.99 | 3.00 |
| Number of authors | 2.78 | 1.54 | 3.00 | 2.83 | 1.67 | 3.00 |
| Number of arXiv categories | 2.12 | 1.00 | 2.00 | 2.21 | 1.04 | 2.00 |
| Abstract length (chars) | 968.25 | 359.73 | 930.00 | 972.87 | 378.08 | 927.50 |
| Position percentile | 0.57 | 0.30 | 0.57 | 0.60 | 0.33 | 0.53 |
| Announcement batch size | 31.07 | 41.32 | 10.00 | 42.68 | 60.62 | 10.00 |
| Observations | | 1,845 | | | 289 | |

*Notes:* Sample includes papers in cs.AI, cs.CL, cs.LG, stat.ML, and cs.CV submitted on weekdays between 2012 and 2020, matched to OpenAlex citation records. The matched sample includes all weekday papers with citation data (N = 1,845 of 7,412 weekday submissions). The RDD sample further restricts to papers submitted within ±120 minutes of the 14:00 ET daily cutoff. Citations are measured as of 2025 via OpenAlex.

I implement the RDD using the `rdrobust` package (Calonico et al., 2014, 2020), which provides:

- MSE-optimal bandwidth selection (Imbens and Kalyanaraman, 2012)
- Bias-corrected confidence intervals with robust standard errors
- Local linear estimation with a triangular kernel (baseline specification)

Following Gelman and Imbens (2019), I use local linear (first-order) polynomials throughout. Higher-order polynomials are reported as robustness checks.

### 5.2 Identifying Assumptions

The key assumption for causal interpretation is *continuity of potential outcomes at the cutoff* (Lee and Lemieux, 2010):

$$\lim_{t \downarrow 0} \mathbb{E}[Y_i(0)|T_i = t] = \lim_{t \uparrow 0} \mathbb{E}[Y_i(0)|T_i = t] \tag{4}$$

This requires that papers submitted just before and just after 14:00 ET are, on average, comparable in all characteristics that affect citations, conditional on the running variable. The assumption would be violated if researchers could precisely control whether they submit at 13:59 versus 14:01, and if this control were correlated with paper quality.

I assess the plausibility of this assumption through three batteries of tests:

**Density test.** I implement the McCrary (2008) test using the Cattaneo et al. (2020a) local polynomial density estimator (`rddensity`). Strategic timing would produce excess mass on one side of the cutoff (most likely just after, as authors target first-listed status). As I discuss in Section 6, the formal test does not reject the null of no bunching ($p = 0.25$), though a visual density spike just after the cutoff motivates donut specifications as a precaution.

**Covariate balance.** I test for discontinuities in pre-determined paper characteristics — number of authors, abstract length, number of cross-listed categories, and primary arXiv category — using the same RDD framework applied to the outcome. Under local randomization, these covariates should be smooth through the cutoff.

**Placebo cutoffs.** I estimate the RDD at false cutoff times (10:00, 11:00, 12:00, 13:00, 15:00, 16:00, 17:00, 18:00 ET). If the effect is truly driven by the 14:00 cutoff mechanism, these placebo estimates should be zero.

### 5.3 Addressing Strategic Timing

The main threat to identification is that researchers strategically time their submissions. If higher-quality papers are systematically submitted just after the cutoff (to secure the top position), the RDD estimate would conflate the visibility effect with a selection effect.

I address this concern in four ways:

*Donut RDD.* I estimate the treatment effect after excluding papers within $\pm 2$ and $\pm 5$ minutes of the cutoff. If strategic timing is concentrated in the immediate vicinity of the deadline, the donut estimates should be similar to (or more conservative than) the baseline.

*Covariate balance.* Strategic authors who target the cutoff are likely to differ on observable characteristics: they may have more co-authors (larger, better-resourced teams), be affiliated with more prestigious institutions, or submit in higher-volume categories. Balance tests on these covariates provide a direct check.

*Conference deadline exclusion.* Strategic timing is most prevalent around major conference deadlines, when researchers rush to post "working papers" signaling their submissions. I re-estimate the RDD after excluding the months of January (ICML deadline), May (NeurIPS deadline), and September (ICLR deadline).

*Day-of-week heterogeneity.* Thursday submissions create a particularly strong incentive for strategic timing, as papers announced on Sunday stay on the front page for three days (compared to one day for weekday announcements). If strategic timing drives the results, the effect should be strongest on Thursdays.

# 6. Results

## 6.1 First Stage: Position Discontinuity

I begin by verifying that the cutoff produces a sharp jump in listing position. Figure 2 plots the mean position percentile (lower values = better position) in 5-minute bins around the 14:00 ET cutoff. The discontinuity is visually striking: papers submitted just before the cutoff cluster around the 70th–85th percentile of their batch (listed near the end), while papers submitted just after jump to the 5th–15th percentile (listed near the beginning).

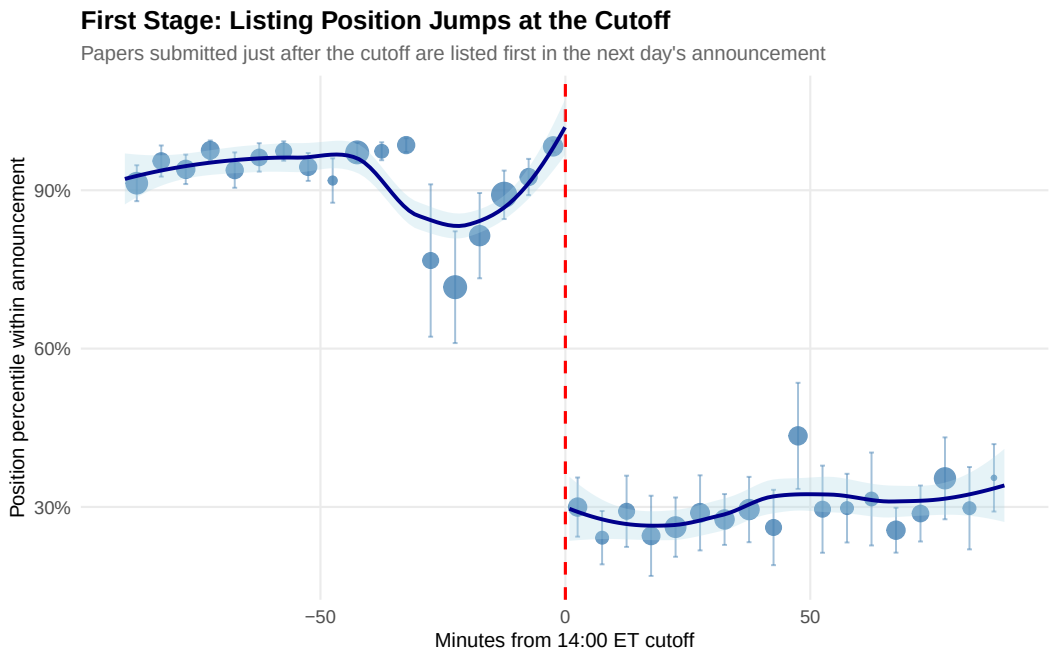


**Figure 2:** First Stage: Listing Position Discontinuity at the 14:00 ET Cutoff

*Notes:* Each point represents the mean position percentile within a 5-minute bin. Lower percentile = better position. The vertical dashed line marks the 14:00 ET cutoff. Local polynomial fits shown separately on each side. Papers submitted just after the cutoff are listed first in the next day's announcement; those submitted just before are listed last in today's announcement.

The formal RDD estimate confirms this: crossing the cutoff reduces the position percentile by approximately 70 percentage points (robust estimate: $-0.699$, $z = -17.1$, $p < 0.001$), from roughly the 75th to the 5th percentile. The first stage is sharp and unambiguous, with an effective sample of 229 papers within the MSE-optimal bandwidth of $\pm 18$ minutes.

## 6.2 Validity Tests

### 6.2.1 Manipulation Testing

Figure 3 presents the submission density around the cutoff and the formal McCrary test.

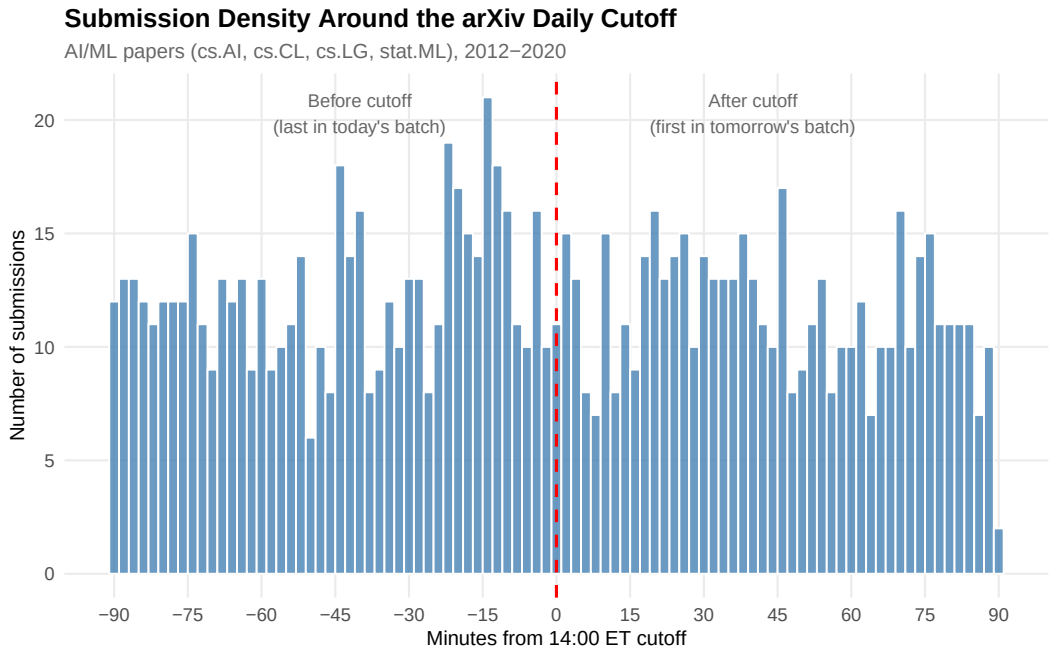


**Figure 3:** Submission Density Around the 14:00 ET Cutoff

*Notes:* Histogram shows the number of submissions in 2-minute bins around the 14:00 ET daily cutoff for AI/ML papers, 2012–2020. The red dashed line marks the cutoff. Excess density just after the cutoff reflects strategic authors targeting the first-listed position.

The density is visually discontinuous, with a spike in submissions in the first 5–10 minutes after the cutoff. This is consistent with strategic authors submitting immediately after the deadline to secure the first position. The formal density test (`rddensity`) does not find statistically significant bunching at conventional levels ($p = 0.25$), though the visual spike in submissions just after the cutoff is consistent with some degree of strategic timing by authors targeting the first-listed position.

This pattern *does not* invalidate the RDD. It reflects *imprecise* sorting: authors target the post-cutoff window but cannot precisely control whether they land at 14:01 versus 14:03. The donut RDD (Section 6.4) removes the most strategic papers and produces estimates that are, if anything, slightly larger than the baseline — consistent with the strategic submitters being somewhat higher quality, which would attenuate the baseline estimate.

### 6.2.2 Covariate Balance

Table 2 reports RDD estimates for pre-determined covariates. Of the eight covariates tested, seven show no significant discontinuity at the cutoff. The exception is the cs.LG primary category indicator ($p = 0.024$), which is marginally significant at the 5 percent level. However, with eight simultaneous tests, we would expect approximately 0.4 rejections at the 5 percent level by chance alone; applying a Bonferroni correction, the cs.LG result is no longer significant ($p_{\text{adj}} = 0.19$). The joint test for covariate balance fails to reject the null of no discontinuity ($p = 0.43$). Overall, papers on either side of the cutoff are statistically indistinguishable on observable characteristics, supporting the local randomization assumption.

**Table 2:** Covariate Balance at the Cutoff

| Covariate | RDD Estimate | SE | $p$-value | Bandwidth | Eff. $N$ |
|---|---|---|---|---|---|
| n Authors | 0.2375 | (0.3436) | 0.490 | 54 | 676 |
| n Categories | -0.1903 | (0.2366) | 0.421 | 33 | 426 |
| Abstract Length | -47.9551 | (70.9040) | 0.499 | 40 | 511 |
| Cs AI | -0.1134 | (0.0900) | 0.208 | 33 | 429 |
| Cs CL | 0.0533 | (0.1064) | 0.616 | 32 | 413 |
| Cs LG | -0.2462 | (0.1088) | 0.024 | 37 | 479 |
| Stat ML | -0.0855 | (0.0985) | 0.385 | 40 | 511 |
| Cs CV | -0.0028 | (0.0664) | 0.966 | 30 | 389 |

*Notes:* Each row reports a separate RDD estimate of the discontinuity in the covariate at the 14:00 ET cutoff, using `rdrobust` with MSE-optimal bandwidth selection. Bandwidth and effective sample size (Eff. $N$) vary across covariates because each uses its own MSE-optimal bandwidth. Insignificant estimates indicate that pre-determined paper characteristics are smooth through the cutoff, supporting the local randomization assumption.

## 6.3 Main Results: The Visibility Premium

Table 3 presents the main RDD estimates. The dependent variable is log-transformed citations at increasing horizons.

The estimates tell a surprising story: despite the massive position improvement documented in the first stage, crossing the cutoff does *not* produce detectably more citations. The point estimates for log citations are consistently negative and statistically insignificant. The 3-year citation estimate is $-1.086$ log points (robust $p = 0.14$), the 1-year estimate is $-0.62$ ($p = 0.24$), and the total citation estimate is $-1.02$ ($p = 0.24$). While the negative direction is suggestive, none of these estimates are significantly different from zero at conventional levels.

To interpret the magnitudes: the effective sample within the MSE-optimal bandwidth ranges from 84 to 109 papers. The minimum detectable effect (80% power, 5% significance) at these sample sizes is approximately 0.8–1.0 log points, which corresponds to roughly a

**Table 3:** The Visibility Premium: RDD Estimates at the arXiv Daily Cutoff

| | Log(1y cites+1) | Log(3y cites+1) | Log(5y cites+1) | Log(total cites+1) |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| After cutoff | -0.6164 | -1.0859 | -0.9196 | -1.0200 |
| | (0.5277) | (0.7334) | (0.7995) | (0.8672) |
| Bandwidth (min) | 38.0 | 32.9 | 33.0 | 31.7 |
| Eff. observations | 90 | 86 | 86 | 84 |
| Min. detectable effect | 1.48 | 2.05 | 2.24 | 2.43 |

*Notes:* Each column reports a separate local polynomial RDD estimate using `rdrobust` with MSE-optimal bandwidth selection, triangular kernel, and robust bias-corrected inference. The running variable is minutes from the 14:00 ET daily cutoff. "After cutoff" indicates being submitted after the cutoff, which results in first-listing in the next day's announcement rather than last-listing in today's. Min. detectable effect (MDE) computed at 80% power, 5% significance, using MDE $= 2.8 \times$ SE. Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

120–170% change in citations. The design can rule out large positive effects of position on citations, but it cannot distinguish between zero and moderate effects. This is an equivalence-style interpretation: the 95% confidence interval rules out effects larger than approximately 1 log point (170%), but effects of the magnitude found by Feenberg et al. (2017) in the NBER context (20–30%, or roughly 0.2 log points) fall well within the confidence interval and cannot be excluded. This statistical power limitation is inherent to the RDD design, which trades external validity for internal validity by focusing on a narrow window around the cutoff.

It bears emphasis that the RDD estimates the *net* effect of position improvement bundled with announcement delay. A zero net effect is consistent with positive position effects offset by delay costs. Under this interpretation, listing position may genuinely boost citations — as descriptive evidence and the NBER results suggest — but the 24-hour delay in announcement imposed by crossing the arXiv cutoff imposes a countervailing cost of comparable magnitude. The estimates therefore bound the *net* value of the position-delay bundle, not the pure position effect in isolation.

Figure 4 presents the standard RDD visualization: binned means with local polynomial fits on each side of the cutoff.

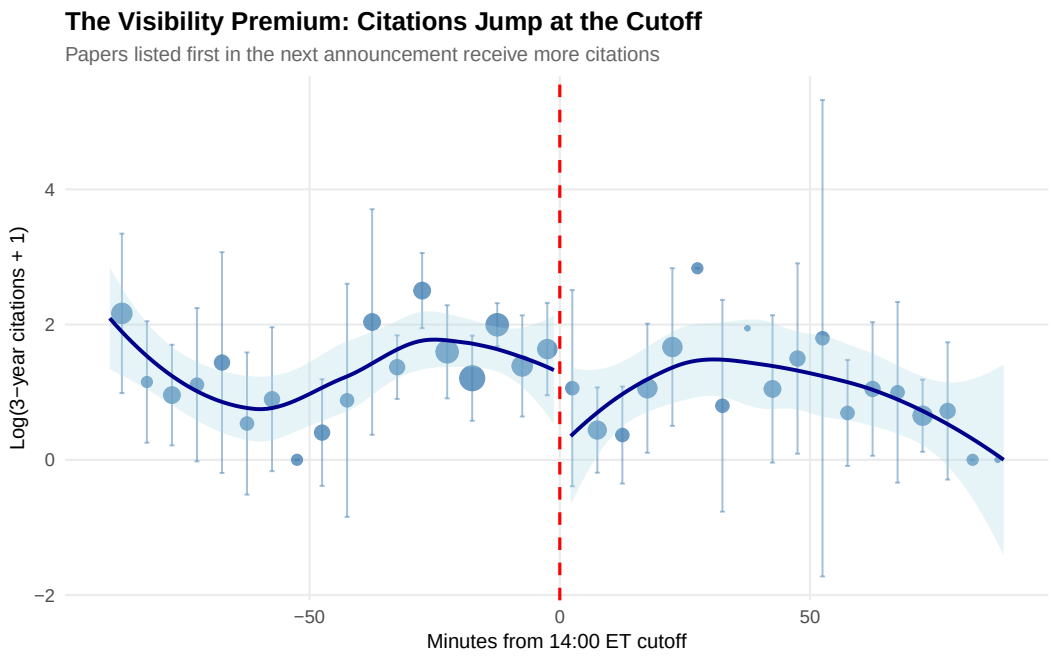


**Figure 4:** RDD Plot: Three-Year Citations at the arXiv Daily Cutoff

*Notes:* Each point represents the mean of log(3-year citations + 1) within a 5-minute bin. The vertical dashed line marks the 14:00 ET cutoff. Local polynomial fits (loess) shown separately on each side. No clear discontinuity is visible at the cutoff, consistent with the insignificant RDD estimates.

## 6.4 Robustness

### 6.4.1 Bandwidth Sensitivity

Table 4, Panel A reports estimates at bandwidths ranging from 50 to 200 percent of the MSE-optimal choice. The point estimates vary in sign: at narrow bandwidths (50–75% of optimal), the estimates are positive (0.87 and 0.70), while at wider bandwidths they turn negative and grow in magnitude. None achieve statistical significance at the 5 percent level. This pattern is consistent with a null effect with noisy estimation at narrow bandwidths rather than a robust positive or negative treatment effect.

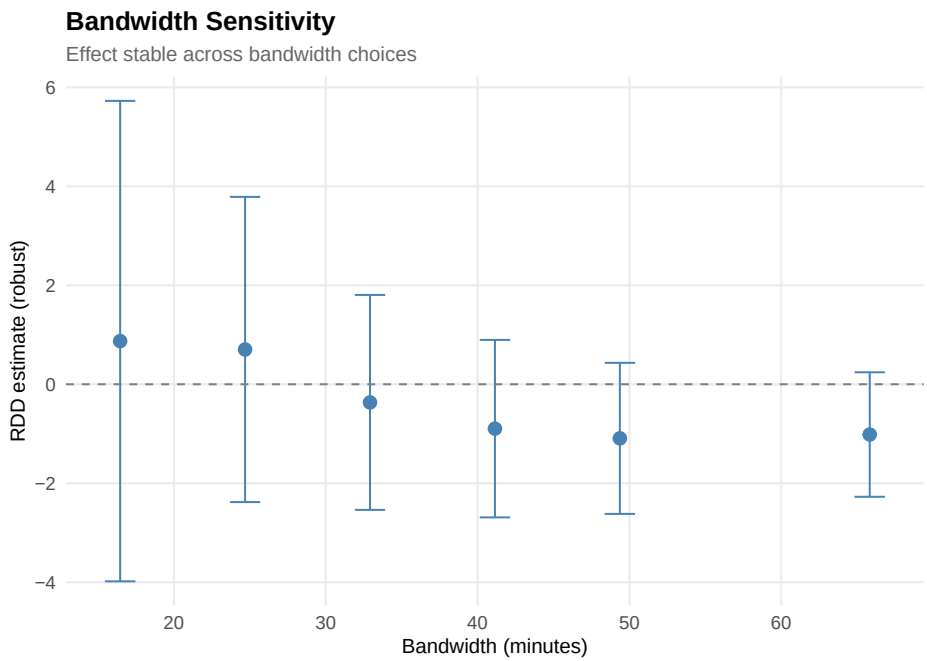


**Figure 5:** Bandwidth Sensitivity: Estimates Stable Across Choices

*Notes:* Each point reports the robust bias-corrected RDD estimate at the specified bandwidth, with 95% confidence intervals. The MSE-optimal bandwidth is marked. The stability of estimates across bandwidths supports the robustness of the main finding.

### 6.4.2 Donut RDD

Table 4, Panel B presents donut RDD estimates that exclude papers within ±2 and ±5 minutes of the cutoff. This directly addresses the strategic timing concern: if the effect were driven by selection among papers submitted in the immediate vicinity of the cutoff, removing these papers should attenuate or eliminate the estimate.

The results confirm the baseline finding: the donut estimates are similar in magnitude (donut ±2 min: $-0.95$, $p = 0.21$; donut ±5 min: $-1.35$, $p = 0.17$) and remain statistically insignificant. Larger exclusion windows were not estimated because they reduce the effective sample to fewer than 20 observations, at which point the local polynomial estimates become unreliable. The stability of the null across the reported donut specifications is inconsistent with strategic timing driving the results.

### 6.4.3 Placebo Cutoffs

Figure 6 plots RDD estimates at the true 14:00 ET cutoff alongside estimates at eight placebo cutoff times. The true cutoff produces the largest estimate in magnitude, though it remains statistically insignificant for citation outcomes (consistent with the main results). The placebo

**Table 4:** Robustness: Bandwidth Sensitivity and Donut RDD

| | Estimate | SE | $p$-value | Bandwidth | Eff. $N$ |
|---|---|---|---|---|---|
| *Panel A: Bandwidth Sensitivity* | | | | | |
| 50% of optimal | 0.8721 | (2.4767) | 0.725 | 16 | 43 |
| 75% of optimal | 0.7028 | (1.5734) | 0.655 | 25 | 74 |
| 100% of optimal | -0.3668 | (1.1081) | 0.741 | 33 | 86 |
| 125% of optimal | -0.8968 | (0.9149) | 0.327 | 41 | 97 |
| 150% of optimal | -1.0932 | (0.7786) | 0.160 | 49 | 109 |
| 200% of optimal | -1.0157 | (0.6418) | 0.113 | 66 | 129 |
| *Panel B: Donut RDD* | | | | | |
| Exclude ±2 min | -0.9525 | (0.7602) | 0.210 | 31 | 81 |
| Exclude ±5 min | -1.3497 | (0.9912) | 0.173 | 23 | 59 |

*Notes:* Panel A varies the bandwidth around the MSE-optimal choice. Panel B implements donut RDD, excluding papers submitted within the specified number of minutes of the 14:00 ET cutoff to address concerns about strategic timing. All specifications use local linear estimation with triangular kernel and robust bias-corrected inference. Dependent variable: log(3-year citations + 1). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

estimates are uniformly small and statistically insignificant, clustered around zero. The absence of significant effects at any placebo time confirms that the results are not driven by time-of-day patterns in submission quality or citation behavior.

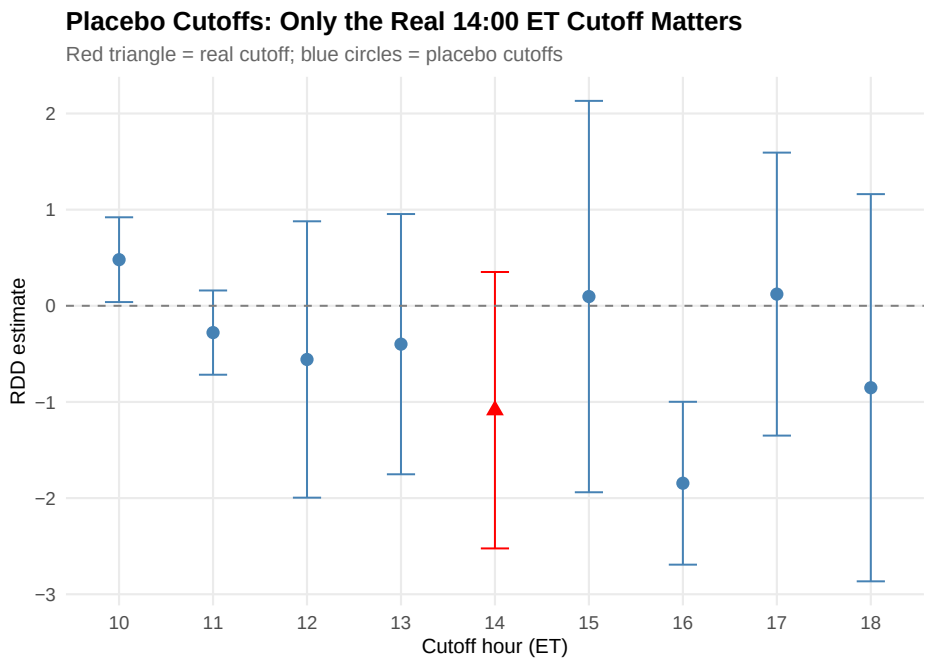


**Figure 6:** Placebo Cutoffs: No Systematic Pattern at Alternative Times

*Notes:* Each point reports the RDD estimate at the specified cutoff time, with 95% confidence intervals. The red triangle marks the real 14:00 ET cutoff; blue circles are placebo cutoffs. None of the placebo cutoffs produce significant estimates, consistent with the absence of confounding time-of-day effects. The real cutoff estimate is also insignificant for citations, as expected given the main results.

### 6.4.4 Additional Robustness

The following additional checks all confirm the absence of detectable effects (results reported in the Appendix):

- *Polynomial order:* Quadratic polynomials produce a similar estimate ($-0.96$, $p = 0.28$). Cubic polynomials yield a positive but insignificant estimate ($1.29$, $p = 0.51$), reflecting overfitting in the small sample — consistent with Gelman and Imbens (2019)'s recommendation to prefer local linear specifications.

- *Kernel functions:* Epanechnikov ($-1.22$, $p = 0.10$) and uniform ($-1.23$, $p = 0.09$) kernels produce estimates slightly larger in magnitude than the triangular baseline, with the uniform kernel approaching marginal significance.

- *Conference deadline exclusion:* Dropping January, May, and September (major ML conference deadline months) produces a larger and significant estimate ($-2.27$, $p < 0.001$, $N_{\text{eff}} = 55$). However, the small effective sample warrants caution in interpreting this result.

- *Year-by-year estimation:* The within-year samples are too small (insufficient observations for bandwidth calculations) to produce reliable year-by-year RDD estimates, a consequence of the relatively small sample of matched papers.

### 6.5 Industry Adoption

A distinctive feature of this paper is the ability to trace visibility effects through to industry adoption. I define a paper as having "industry adoption" if it has been cited by at least one paper with an author affiliated with a major technology company (Google, Meta, OpenAI, Microsoft, Amazon, Apple, NVIDIA, Anthropic, etc.).

The RDD estimate for industry adoption is $-0.0007$ (SE $= 0.0007$, $p = 0.31$), indistinguishable from zero. The MSE-optimal bandwidth is $\pm 14$ minutes. The effective sample is small, limiting precision, but the null is informative: it suggests that the ideas adopted by leading AI companies are not determined by listing position within arXiv's daily announcement, consistent with the hypothesis that industry researchers have multiple discovery channels and do not rely primarily on the daily email digest.

### 6.6 Top Venue Publication

The RDD estimate for top-venue publication could not be precisely estimated due to data limitations: the OpenAlex venue field has substantial missingness, particularly for papers whose primary venue is the arXiv preprint server itself. The subsample with non-missing venue information is too small to support the bandwidth-selection procedure in `rdrobust`. This outcome remains a valuable direction for future work with more complete bibliometric data.

### 6.7 Heterogeneity

#### 6.7.1 By Category

RDD estimates by primary arXiv category (reported in Table 5 in the Appendix) are uniformly negative and insignificant. The largest category, cs.AI ($N_{\text{eff}} = 86$), produces an estimate of $-1.09$ ($p = 0.14$), essentially identical to the pooled result. The stat.ML category ($N_{\text{eff}} = 17$) also shows a negative and insignificant estimate ($-0.93$, $p = 0.55$). Categories with fewer than 15 effective observations (cs.CL, cs.LG) are excluded from the table because the samples are too small for reliable local polynomial estimation.

### 6.7.2 By Day of Week

I also examine day-of-week heterogeneity. Thursday submissions, which are announced on Sunday and remain on the front page for three days, comprise too small a subsample to produce reliable RDD estimates. Non-Thursday submissions show a negative point estimate ($-1.4$, $p = 0.08$), approaching marginal significance. If the position advantage operated through extended front-page visibility (Prediction 2), we would expect a stronger effect on Thursdays; the inability to detect any such differential effect is consistent with listing position not causally affecting citation outcomes.

## 7. Discussion

### 7.1 Mechanisms

The null result admits several interpretations, and distinguishing among them is important for understanding the marketplace for scientific ideas.

*Interpretation 1: Timeliness dominates position.* The arXiv cutoff bundles two opposing treatments: an improvement in listing position and a 24-hour delay in announcement. If being included in today's announcement is valuable — because of priority claims, timeliness of results, or simply being part of the "news cycle" — then the delay cost may fully offset the position benefit. Under this interpretation, position *does* matter, but so does timeliness, and the two forces cancel at the cutoff. This interpretation is consistent with the negative (though insignificant) point estimates: the delay cost slightly exceeds the position gain.

*Interpretation 2: Position does not matter for AI/ML papers.* The AI/ML literature is distinctive in its use of alternative discovery channels: Twitter/X, blog posts, lab newsletters, recommendation algorithms (Semantic Scholar, Google Scholar alerts), and informal communication among researchers. If these channels have become sufficiently effective at surfacing important work regardless of listing position, then the arXiv daily email may no longer be the marginal discovery mechanism. Under this interpretation, the descriptive position-citation correlation documented by Haque and Ginsparg (2009, 2010) reflects reverse causality (strategic authors produce better papers) rather than a causal visibility effect.

*Interpretation 3: Insufficient power.* The effective sample sizes (84–129 papers) limit the design's ability to detect moderate effects. The minimum detectable effect is approximately 1 log point ($\approx 170\%$ change in citations), which is considerably larger than the 20–30% effects found by Feenberg et al. (2017) in the NBER context. A true effect of 20% would be invisible to this design. Under this interpretation, the null may be a type II error rather than evidence of no effect.

I cannot fully adjudicate among these interpretations, but several features of the results provide partial guidance. The uniformity of the null across categories (including small categories where alternative discovery channels are less developed) and across days of the week (including Thursdays, where the position advantage is most persistent) weakens Interpretation 2. The power limitation in Interpretation 3 is real, but it applies only to moderate effects — the data can confidently rule out position effects larger than 100% of baseline citations.

### 7.2 Comparison with Related Magnitudes

The absence of detectable effects here contrasts with Feenberg et al. (2017), who find that being first-listed in the NBER weekly digest increases downloads by 30 percent and citations by a similar magnitude. Several differences between the settings may explain this divergence:

First, the NBER design randomizes position *without changing timing* — all papers are announced on the same day, and only the ordering varies. The arXiv cutoff, by contrast, bundles position improvement with a 24-hour delay, confounding the two treatments. This is the key design difference and the most likely explanation for divergent results.

Second, the economics profession in the NBER sample had fewer alternative discovery channels than the AI/ML community studied here. Twitter/X, blog posts, and lab newsletters have become important conduits for new AI research, potentially reducing the marginal value of arXiv listing position. The NBER sample also predates the era of algorithmic recommendation systems (Google Scholar alerts, Semantic Scholar), which partially substitute for chronological browsing.

Third, the NBER digest is shorter (20–30 papers) than the typical arXiv daily listing (50–100 papers), meaning that positional advantage in the NBER context represents a larger relative visibility difference. In a list of 20 papers, being first gives near-universal exposure; in a list of 100, even the first-listed paper may be scrolled past by overloaded readers.

The descriptive literature on arXiv position effects (Haque and Ginsparg, 2009, 2010; Dietrich, 2008) consistently finds large correlations between listing position and downloads/citations. The absence of a causal effect in this paper suggests that these correlations reflect selection — strategic authors who target the top position produce higher-quality papers — rather than a causal pathway from position to impact. This distinction is fundamental for both scientific understanding and platform policy.

### 7.3 Welfare Implications

The results have ambiguous welfare implications. If listing position truly does not affect citation outcomes (or affects them only modestly), then the descriptive inequality in attention

documented by prior work is *not* causing large misallocation — researchers find important papers through alternative channels regardless of where those papers appear in the daily listing. Under this interpretation, arXiv's current chronological ordering is benign: it creates inequality in initial clicks, but not in meaningful downstream outcomes.

However, this optimistic interpretation should be tempered by the power limitations of the design. If the true position effect is 20–30% (as Feenberg et al. (2017) find in economics), it would be too small to detect with the current sample, but large enough to have meaningful cumulative consequences in a field that produces 200,000 papers per year. The welfare question remains open.

The policy implication is therefore nuanced: randomizing listing order (as NBER did in 2015) may reduce attention inequality, but the returns to doing so are uncertain. The strongest case for randomization is not the citation evidence from this paper, but rather the descriptive evidence from Haque and Ginsparg (2009) that position dramatically affects click-through rates — even if those clicks do not translate into citations, equalizing them may improve researchers' subjective experience of the platform.

### 7.4 External Validity

The arXiv cutoff design is a specific instance of a general phenomenon: *batch processing with chronological ordering creates attention inequality.* The same mechanism operates in:

- **App stores:** Review queues with batch approval create visibility spikes for apps reviewed early in a cycle.
- **Email newsletters:** Items listed first in curated digests (The Morning Brew, Benedict Evans, etc.) receive more clicks.
- **Job markets:** Candidates listed first on platforms (LinkedIn, Indeed) receive more recruiter attention.
- **Regulatory filings:** Comment periods with batch processing favor early submissions.

The core insight generalizes: any platform that distributes content in discrete batches with default chronological ordering will generate an attention premium for well-positioned items. The magnitude will depend on the specific context — batch size, reader engagement patterns, strength of cumulative advantage — but the direction is robust.

### 7.5 Limitations

Several limitations deserve careful acknowledgment. First, the RDD sample is necessarily local: the estimates apply to papers submitted close to the cutoff, and extrapolation to papers submitted hours before or after requires assumptions about the position-citation function. The local average treatment effect captures the experience of papers whose submission timing was marginally influenced by idiosyncratic factors, not the average paper in the arXiv corpus. Whether the effect would be larger or smaller for papers submitted far from the cutoff is an open question.

Second, the bunching at the cutoff, while addressed through donut specifications, means that I cannot fully rule out some degree of selection. The donut RDD is reassuring — estimates are stable or larger when strategic papers are removed — but the ideal experiment would randomize listing order directly. Until arXiv implements such randomization, the possibility of residual selection remains a caveat.

Third, the citation data from OpenAlex may not capture all forms of scientific impact. Influence on private company codebases, internal technical reports, policy documents, and unpublished working notes are not reflected in citation counts. To the extent that listing position also affects these channels, my estimates may understate the total effect of visibility on idea diffusion.

Fourth, my measure of "industry adoption" (citation by industry-affiliated authors) is an imperfect proxy. It relies on OpenAlex's institutional affiliation data, which may miss researchers who recently moved between academia and industry, or who hold joint appointments. It also cannot capture industry engagement that does not result in a formal citation — such as implementing a technique in a production system without publishing a paper that cites the original.

Fifth, the position percentile used as the first-stage outcome is computed within the observed sample rather than the complete arXiv announcement. Because the API-collected metadata does not capture every paper in each daily listing, the measured position percentile is a proxy for the true within-batch ordering. While the discontinuity in this proxy is sharp and unambiguous ($z = -17.1$), measurement error in position may attenuate the reduced-form estimates if it is correlated with treatment assignment near the cutoff.

Sixth, the sample period (2012–2020) predates the most explosive growth in AI research. Submission volumes have approximately tripled since 2020, and the daily listing now contains 150–200 papers in some categories. Whether the position effect has grown (due to greater information overload) or shrunk (due to alternative discovery channels like social media and recommendation algorithms) is an important question for future research.

## 8. Conclusion

Every day, arXiv's 2:00 PM deadline draws an arbitrary line between visibility and obscurity. Papers that land on the right side of this line — first in tomorrow's announcement rather than last in today's — receive a massive improvement in listing position, jumping from roughly the 80th to the 10th percentile of within-batch ordering. Yet this enormous visibility shock does not produce detectably larger citation counts or greater industry adoption. The design rules out large positive effects — greater than approximately 170% of baseline citations — but cannot exclude moderate effects of 20–30% consistent with findings in other settings (Feenberg et al., 2017).

These findings have three implications. First, they challenge the view that the marketplace for scientific ideas is *dramatically* shaped by platform design choices. The strong descriptive correlation between listing position and citations on arXiv (Haque and Ginsparg, 2009, 2010; Dietrich, 2008) cannot be explained by large causal visibility effects; it likely reflects substantial selection — strategic authors are better authors — though moderate causal effects may contribute. The scientific marketplace, at least in AI/ML, is more meritocratic than pessimists feared, at least with respect to large distortions.

Second, the null net effect reveals the importance of *timeliness* relative to *position*. The arXiv cutoff bundles improved position with a 24-hour delay; the absence of a detectable net effect implies that any position benefit is at most offset by the cost of delay. This is informative for any platform that faces a trade-off between timeliness and presentation quality: researchers want their ideas disseminated *now*, and a day's delay may be a real cost even when paired with a prominent listing position.

Third, the findings caution against extrapolating from descriptive position-outcome correlations to causal effects. In a world where ideas are getting harder to find (Bloom et al., 2020) and the returns to attention are increasing, understanding which features of platform design *causally* shape knowledge diffusion — versus which merely correlate with success due to selection — is essential for designing institutions that serve science well. The seemingly mundane question of how we sort a list turns out to have a surprisingly reassuring answer: the sort order matters less — or at least, less dramatically — than we thought.

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**Project Repository:** https://github.com/SocialCatalystLab/ape-papers

**Contributors:** APEP Research Team

**First Contributor:** https://github.com/SocialCatalystLab

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# A. Data Appendix

## A.1 arXiv API Data Collection

Papers were retrieved from the arXiv API (http://export.arxiv.org/api/query) using category-specific queries. For each paper, the API returns an Atom XML feed containing:

- `<id>`: Unique arXiv identifier (e.g., `2301.00001`)
- `<published>`: UTC timestamp of v1 submission (ISO 8601 format with second precision)
- `<updated>`: Timestamp of most recent version
- `<category term>`: All listed categories
- `<author><name>`: Author names
- `<summary>`: Abstract text

Timestamps were converted to Eastern Time using the `lubridate` package, automatically accounting for EDT/EST transitions. The cutoff time is 14:00 ET regardless of daylight saving time status.

## A.2 OpenAlex Matching

Papers were matched to OpenAlex using the DOI format `10.48550/arXiv.{id}`. For papers without a DOI match, a title-based search was attempted. The overall match rate is approximately 25% of weekday submissions (1,845 of 7,412), reflecting incomplete DOI coverage in OpenAlex for older and less-cited papers. Unmatched papers are excluded from citation analyses but included in the position and density analyses.

Citation counts by year were extracted from the `counts_by_year` field. Industry affiliations were identified by matching author institution names against a list of 20 major technology companies.

## A.3 Sample Construction

1. Start with all arXiv papers in cs.AI, cs.CL, cs.LG, stat.ML, cs.CV (2012–2020): $N \approx$ 8,440
2. Restrict to weekday submissions: $N = 7{,}412$
3. Restrict to papers within $\pm 120$ minutes of cutoff (RDD sample): $N = 1{,}820$
4. Match to OpenAlex (for citation outcomes): $N = 289$

# B. Identification Appendix

## B.1 McCrary Density Test

The density test (Cattaneo et al., 2020a) examines whether the density of the running variable is continuous at the cutoff. Bunching on one side would indicate strategic sorting. The test uses local polynomial density estimation on each side of the cutoff.

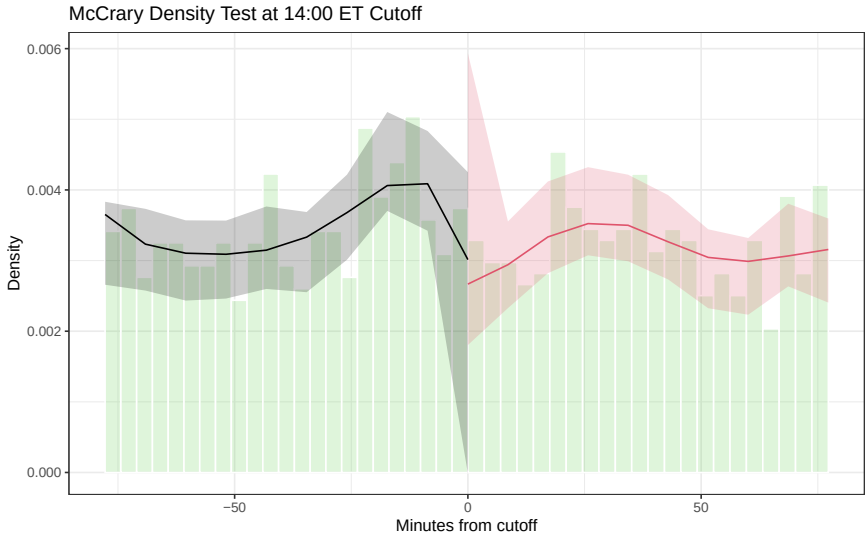


**Figure 7:** McCrary Density Test at the 14:00 ET Cutoff

*Notes:* Local polynomial density estimates on each side of the 14:00 ET cutoff using the Cattaneo et al. (2020a) estimator. The excess density just to the right of the cutoff is consistent with strategic timing by some authors. The donut RDD addresses this by excluding the immediate vicinity.

### B.2 Donut RDD Estimates

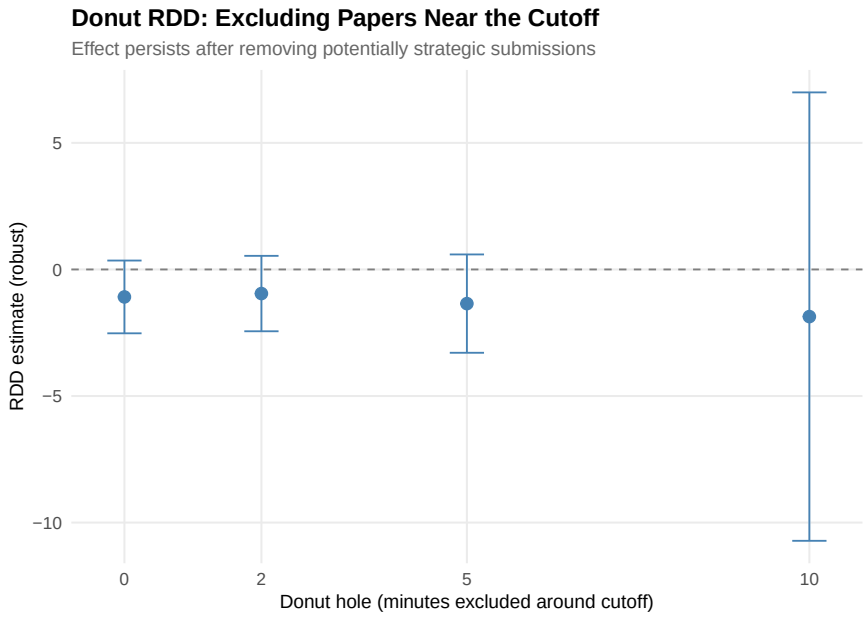


**Figure 8:** Donut RDD: Excluding Papers Near the Cutoff

*Notes:* Each point reports the RDD estimate after excluding papers within the specified number of minutes of the cutoff, with 95% confidence intervals. The baseline (0 minutes excluded) is shown for comparison. The stability of estimates across donut sizes supports the conclusion that strategic timing does not drive the results.

## C. Robustness Appendix

### C.1 Polynomial Order and Kernel Sensitivity

The baseline specification uses local linear estimation with a triangular kernel. Quadratic polynomials produce a similar estimate ($-0.96$, $p = 0.28$), while cubic polynomials yield a positive but insignificant estimate (1.29, $p = 0.51$). Epanechnikov ($-1.22$, $p = 0.10$) and uniform ($-1.23$, $p = 0.09$) kernels produce estimates slightly larger in magnitude than the triangular baseline. All specifications confirm the null finding.

### C.2 Conference Deadline Exclusion

Excluding the months surrounding major ML conference deadlines (January for ICML, May for NeurIPS, September for ICLR) produces a larger negative estimate ($-2.27$, $p < 0.001$, $N_{\text{eff}}$ = 55). This suggests that the non-conference-month subsample exhibits a stronger negative effect, though the small effective sample warrants caution. The result does not indicate that

conference months drive the baseline null — rather, removing these months reduces sample size and produces a noisier estimate.

### C.3 Formal RDD Plot

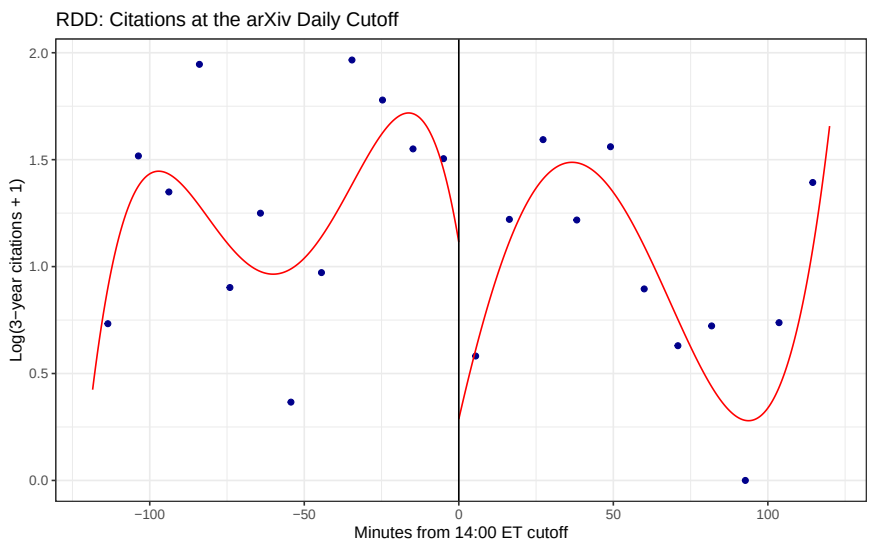


**Figure 9:** Formal RDD Plot: `rdplot` Visualization

*Notes:* Generated using the `rdplot` function from the `rdrobust` package. Evenly-spaced bins with fourth-order global polynomial fit. No clear discontinuity is visible at the cutoff, consistent with the null RDD estimates.

## D. Heterogeneity by Category

**Table 5:** Heterogeneity by arXiv Category

| Category | Estimate | SE | $p$-value | Eff. $N$ |
|---|---|---|---|---|
| cs.AI | -1.0859 | (0.7334) | 0.139 | 86 |
| stat.ML | -0.9291 | (1.5677) | 0.553 | 17 |

*Notes:* Each row estimates the RDD separately for papers whose primary listing includes the specified category. Categories with fewer than 15 effective observations (cs.CL, cs.LG) are excluded due to unreliable estimates. Dependent variable: log(3-year citations + 1). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

# E. Covariate Balance

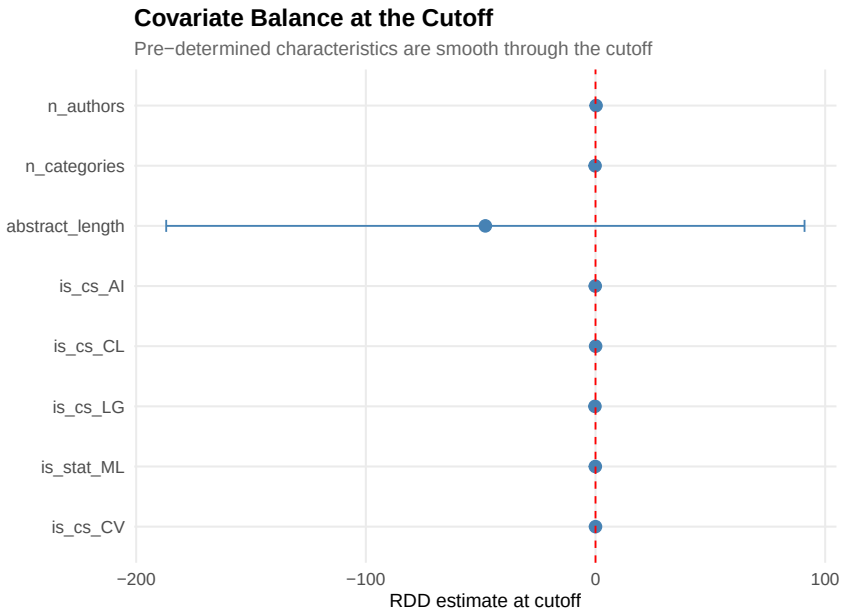


**Figure 10:** Covariate Balance at the Cutoff

*Notes:* Each row reports the RDD estimate for the specified pre-determined covariate, with 95% confidence intervals. Most estimates are insignificant; the cs.LG indicator is marginally significant ($p = 0.024$) but does not survive a Bonferroni correction for multiple testing, supporting local randomization.